

AN INFINITE DENSE COUNTEREXAMPLE FAMILY FOR EXTREMAL FIRST BETTI NUMBERS OF FLAG COMPLEXES

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ABSTRACT. Beers and Bakke Botnan conjectured that every graph maximizing the first reduced Betti number of its flag complex among graphs with fixed numbers of vertices and edges contains a complete bipartite spanning subgraph. We give an infinite family of counterexamples strictly above the Turán threshold that motivates the conjecture. For every $n \geq 7$, let H_n consist of a triangle, a two-edge path attached to one triangle vertex, and $n - 5$ leaves at the other end of the path, and put $G_n = \overline{H_n}$. Then G_n has $\binom{n}{2} - n > \lfloor n^2/4 \rfloor$ edges, its flag complex is homotopy equivalent to $S^1 \vee S^1$, and 2 is the maximum first reduced Betti number among all graphs with the same numbers of vertices and edges. Since H_n is connected, G_n has no complete bipartite spanning subgraph. The proof reduces the extremal upper bound to the independence complexes of graphs with average degree two and then uses a leaf reduction and the homotopy types of cycle independence complexes. Complete exact censuses at $(7, 14)$ and $(8, 20)$, including two independent labeled implementations, verify the distributions of Betti numbers and identify the violating maximizers by explicit permutation-orbit equality.

1. INTRODUCTION

For a finite simple graph G , write $\text{Cl}(G)$ for its flag, or clique, complex and set

$$\tilde{\beta}_1(\text{Cl}(G); \mathbb{k}) := \dim_{\mathbb{k}} \tilde{H}_1(\text{Cl}(G); \mathbb{k}),$$

where $\mathbb{k}$ is an arbitrary field. Beers and Bakke Botnan study extremal Betti numbers at fixed order and size and conjecture that every maximizer contains a complete bipartite spanning subgraph [2, Conjecture 36]. The displayed conjecture in the full preprint is Conjecture 3. Its surrounding examples and plots emphasize the dense range $m \geq \lfloor n^2/4 \rfloor$.

As written, however, the conjecture has no density hypothesis. Sparse counterexamples are elementary: C_5 is maximizing at $(n, m) = (5, 5)$ and has connected complement, and an exact order-five census finds the still earlier size-4 example $C_4 \sqcup K_1$. The contribution here is therefore not the first literal counterexample. It is the persistence of the obstruction in an

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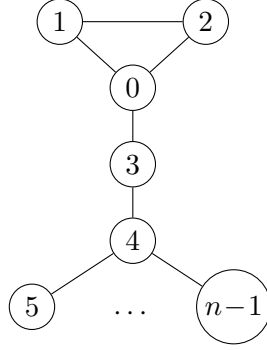


FIGURE 1. The connected complement $H_n = \overline{G_n}$. Every pair not shown as an edge is an edge of G_n .

infinite family strictly above the Turán threshold, where the source paper explicitly studies the fixed-size extremal problem.

We disprove the conjecture throughout an infinite sequence in that dense range. The construction is easiest to see in the complement. For $n \geq 7$, let H_n be the graph on $\{0, 1, \dots, n-1\}$ with edge set

$$(1) \quad E(H_n) = \{01, 12, 20, 03, 34\} \cup \{4j : 5 \leq j \leq n-1\},$$

and define $G_n = \overline{H_n}$. Thus H_n contains a triangle on $\{0, 1, 2\}$, the path 0–3–4, and $n-5$ leaves adjacent to 4.

Theorem 1.1. *For every integer $n \geq 7$ and every field $\mathbb{k}$, the graph G_n defined by (1) satisfies*

$$|V(G_n)| = n, \quad |E(G_n)| = \binom{n}{2} - n,$$

and

$$\tilde{\beta}_1(\text{Cl}(G_n); \mathbb{k}) = 2 = \max_{\substack{|V(F)|=n \\ |E(F)|=\binom{n}{2}-n}} \tilde{\beta}_1(\text{Cl}(F); \mathbb{k}).$$

Moreover,

$$\text{Cl}(G_n) \simeq S^1 \vee S^1, \quad \tilde{H}_1(\text{Cl}(G_n); \mathbb{Z}) \cong \mathbb{Z}^2.$$

Nevertheless, G_n contains no complete bipartite spanning subgraph. Consequently, the Beers–Bakke Botnan conjecture is false for infinitely many parameter pairs strictly above the Turán threshold.

The argument is field-independent and theoretical. Exact graph censuses and a Lean certificate accompany the manuscript, but neither is a premise of Theorem 1.1. Earlier work on extremal Betti numbers as a function of the number of vertices alone addresses a different optimization problem [1].

2. INDEPENDENCE-COMPLEX TOOLS

For a graph Q , let $\text{Ind}(Q)$ denote its independence complex. A set is a clique in G precisely when it is independent in $\overline{G}$, and hence

$$(2) \quad \text{Cl}(G) = \text{Ind}(\overline{G}).$$

We use two standard facts about independence complexes [3, 4]. The first is included with proof to fix conventions.

Lemma 2.1 (Leaf reduction). *Let v be a leaf of Q , let w be its unique neighbor, and put $J = Q - N_Q[w]$. Then*

$$\text{Ind}(Q) \simeq \Sigma \text{Ind}(J).$$

Consequently, for every field $\mathbb{k}$ and every integer i ,

$$\tilde{H}_i(\text{Ind}(Q); \mathbb{k}) \cong \tilde{H}_{i-1}(\text{Ind}(J); \mathbb{k}).$$

In particular, $\tilde{\beta}_1(\text{Ind}(Q); \mathbb{k}) = \tilde{\beta}_0(\text{Ind}(J); \mathbb{k})$.

Proof. Write

$$A = \text{Ind}(Q - w), \quad B = w * \text{Ind}(J).$$

Every independent set either avoids w and belongs to A , or contains w and belongs to B , so $\text{Ind}(Q) = A \cup B$ and $A \cap B = \text{Ind}(J)$. The complex A is a cone with apex v , because v is isolated in $Q - w$, and B is a cone with apex w . Their inclusions are simplicial cofibrations, so this union is a homotopy pushout of two contractible spaces over $\text{Ind}(J)$ and is homotopy equivalent to $\Sigma \text{Ind}(J)$. The reduced Mayer–Vietoris sequence gives the stated degree shift. $\square$

For disjoint graphs,

$$(3) \quad \text{Ind}(Q_1 \sqcup Q_2) = \text{Ind}(Q_1) * \text{Ind}(Q_2).$$

We also use the standard cycle calculation

$$(4) \quad \text{Ind}(C_\ell) \simeq \begin{cases} S^{r-1} \vee S^{r-1}, & \ell = 3r, \\ S^{r-1}, & \ell = 3r + 1, \\ S^r, & \ell = 3r + 2. \end{cases}$$

The formula follows, for example, from the usual discrete-Morse matching for cycle independence complexes [4, Proposition 11.17].

3. THE EXTREMAL UPPER BOUND

Proposition 3.1. *Let $n \geq 7$. For every graph F on n vertices with $\binom{n}{2} - n$ edges and every field $\mathbb{k}$,*

$$\tilde{\beta}_1(\text{Cl}(F); \mathbb{k}) \leq 2.$$

Proof. Put $Q = \overline{F}$. Then Q has n vertices and exactly n edges, so its average degree is 2 and $\delta(Q) \leq 2$. By (2), it suffices to bound $\tilde{\beta}_1(\text{Ind}(Q); \mathbb{k})$. We distinguish the possible values of $\delta(Q)$.

If $\delta(Q) = 0$, an isolated vertex of Q is a cone point of $\text{Ind}(Q)$, so all reduced homology vanishes.

Suppose $\delta(Q) = 1$. Choose a leaf v with neighbor w , write $d = \deg_Q(w)$, and put $J = Q - N_Q[w]$. With $N = |V(J)| = n - d - 1$, deletion of $N_Q[w]$ removes at least the d edges incident with w , and therefore

$$(5) \quad |E(J)| \leq n - d = N + 1.$$

By Lemma 2.1,

$$\tilde{\beta}_1(\text{Ind}(Q); \mathbb{k}) = \tilde{\beta}_0(\text{Ind}(J); \mathbb{k}).$$

The connected components of $\text{Ind}(J)$ are those of its one-skeleton $\overline{J}$. If $\overline{J}$ had at least four components with nonempty vertex classes of sizes $a_1, \dots, a_s$, then J would contain every edge between distinct classes. Merge classes until four remain; merging only removes cross edges. Write the four resulting sizes as $1 + x_i$, where $x_i \geq 0$ and $\sum_i x_i = N - 4$. Their number of cross edges is

$$\sum_{i < j} (1 + x_i)(1 + x_j) = 3N - 6 + \sum_{i < j} x_i x_j \geq 3N - 6.$$

For $N \geq 4$, this is greater than $N + 1$, contradicting (5). When $N < 4$, four components are impossible. Thus $\text{Ind}(J)$ has at most three components, and its reduced zeroth Betti number is at most 2.

Finally, suppose $\delta(Q) = 2$. Since the average degree is also 2, every vertex has degree 2, so Q is a disjoint union of cycles. If Q is one cycle, (4) shows that for $n \geq 7$ its first Betti number is at most 1. If Q has two cycle components, the join formula (3) and the reduced-homology formula for joins show that first homology can occur only when both factors have reduced homology in degree zero. Formula (4) gives reduced zeroth ranks 2 and 1 for C_3 and C_4 , respectively, and zero for every longer cycle. Thus $C_3 \sqcup C_4$ gives rank 2 and $C_4 \sqcup C_4$ gives rank 1; the only larger product, from $C_3 \sqcup C_3$, has six vertices and is excluded by $n \geq 7$. With three or more cycle components, the join shifts the least possible nonzero reduced-homology degree to at least 2. Hence the first Betti number is again at most 2. $\square$

4. THE COUNTEREXAMPLE FAMILY

Lemma 4.1 (Spanning biclique criterion). *Let G be a finite graph on at least two vertices. Then G contains a complete bipartite spanning subgraph with two nonempty partition classes if and only if $\overline{G}$ is disconnected.*

Proof. If $A \sqcup B = V(G)$ is a nontrivial bipartition and every edge between A and B belongs to G , then $\overline{G}$ has no edge between the two parts and is disconnected. Conversely, if $\overline{G}$ is disconnected, take A to be the vertex set

of one component and $B = V(G) \setminus A$. Both are nonempty, no complement edge joins them, and hence every cross pair is an edge of G . $\square$

Proof of Theorem 1.1. Equation (1) gives

$$|E(H_n)| = 3 + 2 + (n - 5) = n,$$

so $G_n = \overline{H_n}$ has $\binom{n}{2} - n$ edges. The graph H_n is connected. Lemma 4.1 therefore shows that G_n contains no complete bipartite spanning subgraph.

It remains to compute the Betti number. Vertex 5 is a leaf of H_n with neighbor 4, and

$$H_n - N_{H_n}[4] = H_n[\{0, 1, 2\}] = K_3.$$

The complex $\text{Ind}(K_3)$ consists of three isolated vertices and is homotopy equivalent to $S^0 \vee S^0$. Lemma 2.1 and (2) give

$$\text{Cl}(G_n) = \text{Ind}(H_n) \simeq \Sigma \text{Ind}(K_3) \simeq S^1 \vee S^1.$$

Consequently,

$$\tilde{\beta}_1(\text{Cl}(G_n); \mathbb{k}) = 2, \quad \tilde{H}_1(\text{Cl}(G_n); \mathbb{Z}) \cong \mathbb{Z}^2.$$

Proposition 3.1 proves extremality. Finally, $\binom{n}{2} - n > \lfloor n^2/4 \rfloor$ for $n \geq 7$, so every member lies strictly above the Turán threshold. $\square$

The smallest member is isomorphic to the graph with graph6 encoding FEzvg; this identifier is included with the accompanying machine-readable witnesses.

5. INDEPENDENT EXACT VERIFICATION

The accompanying programs independently check the theorem's finite boundary cases. They use exact row reduction and Smith normal form, and they do not call numerical topology software. The complete fixed-pair censuses give the following labeled distributions; the final column counts maximizing graphs with connected complement.

TABLE 1. Complete exact labeled censuses at the first two family parameters.

n	m	$\tilde{\beta}_1 = 0$	$\tilde{\beta}_1 = 1$	$\tilde{\beta}_1 = 2$	connected max.
7	14	82,320	31,545	2,415	1,260
8	20	2,873,430	227,955	6,720	3,360

At (7, 14) there are four unlabeled maximizing classes, exactly one with connected complement. At (8, 20) there are three unlabeled maximizing classes, whose complement structures are listed in Table 2. Here $K_{1,4} + e$ means that an edge is added between two leaves.

The Python labeled verifier enumerates every graph at each of these two parameter pairs. It separately generates every relabeling of G_7 and G_8 ,

TABLE 2. All maximizing isomorphism classes at $(8, 20)$.

Complement representative	labeled orbit	connected
H_8	3,360	yes
$K_3 \sqcup (K_{1,4} + e)$	1,680	no
$(K_4 - e) \sqcup K_{1,3}$	1,680	no

stores the maximizing graphs with connected complement, and checks equality of the two sets. An independently written C implementation repeats the enumeration and the same set comparison. Thus the one-class conclusion does not follow merely from matching the orbit sizes 1,260 or 3,360. A third implementation streams nauty representatives, weights each class by $n!/|\text{Aut}(G)|$, and recovers Table 1 over $\mathbb{Q}, \mathbb{F}_2, \mathbb{F}_3$, and $\mathbb{F}_5$.

Proposition 5.1 (Finite post-Turán minimality). *Among parameter pairs with $n \leq 8$ and $m > \lfloor n^2/4 \rfloor$, a maximizing graph with connected complement exists only at $(7, 14)$ and $(8, 20)$. In particular, $(7, 14)$ is the smallest such failure by order, and then by size. This conclusion holds over every coefficient field.*

Proof. The verification streams one representative of every graph isomorphism class through order eight. For every graph in the strict post-Turán range it presents the triangle-boundary subgroup in a fundamental-cycle basis and computes its Smith normal form. No torsion invariant occurs. The resulting integral free ranks identify exactly the two parameter pairs in the statement; the universal coefficient theorem makes the conclusion field-independent. The complete command, source, and captured output are included in the supplement. $\square$

The all-order proof of Theorem 1.1 does not depend on these enumerations.

The Lean companion verifies the family definition, the exact all-order degree formula and edge counts for H_n and G_n , connectedness of H_n , the generic connected-complement obstruction of Lemma 4.1, the deletion certificate $H_n - N[4] = K_3$, and the elementary dense-range inequalities. It contains no `sorry` or declared axiom. It does not formalize simplicial homology, the leaf reduction, or the cycle formula and is therefore a structural certificate rather than a formalization of the full theorem.

6. DISCUSSION

The family separates the structural conclusion of the conjecture from the value of the extremal invariant. The complements H_n are sparse, connected, and unicyclic, yet their independence complexes attain the largest possible

first Betti number at the corresponding dense edge counts. The extremal upper bound depends only on the elementary fact that every graph of average degree two has a vertex of degree at most two.

The conjecture’s published formulation is universal: it asserts the spanning structure for every maximizing graph. A single maximizing graph with connected complement therefore refutes it. The formulation is false even along a regular dense sequence. A weaker existential question—whether every dense pair (n, m) admits *some* maximizer with a complete bipartite spanning subgraph—is not decided by this family, because other maximizers can coexist at the same pair. The exact census at $(8, 21)$, for example, has 280 labeled maximizers and all of them have disconnected complements.

DATA AND CODE AVAILABILITY

The supplementary archive contains graph6 witnesses, the Python and C labeled censuses, the independent unlabeled and integral audits, the Lean source and pinned dependency files, captured verification data, and SHA-256 checksums. No floating-point computation is used to certify a theorem. The complete versioned preprint and verification artifact is archived at Zenodo: <https://doi.org/10.5281/zenodo.21892997>.

DECLARATION OF AGENTIC AI AND AI-ASSISTED TECHNOLOGIES

During the discovery, verification, and preparation of this work, the author used OpenAI Codex and Anthropic Claude as agentic research assistants. Codex assisted with computational exploration, literature and novelty searches, code and Lean development, proof development, and manuscript drafting and editing. Claude was used for independent adversarial review, including attempts to identify mathematical, computational, citation, and presentation defects. The author independently checked the claims, proofs, computations, sources, and final text, accepts full responsibility for the work, and confirms that neither system is an author.

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